HBM-resident bitstream inspection

FPGA_HLS4ML versus GPU_P2P

Report created: 2026-11-08 · Measurements: 2026-11-08 · Host: hacc-gpu-u55c-01.inf.ethz.ch

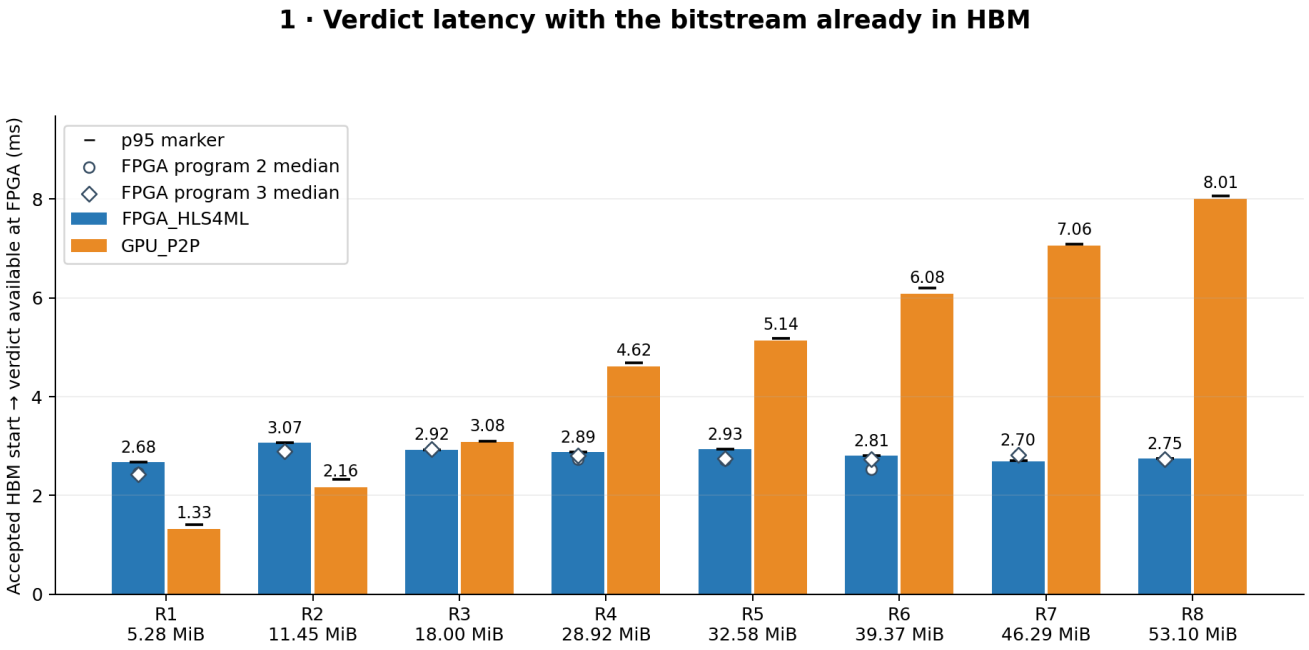
Question: when the complete raw bitstream is already in FPGA HBM, how soon can the FPGA obtain a classification verdict? OPT_10 selectively gathers the sampled bytes and streams them into the resource-optimized CNN. OPT_11 forwards all bytes to MI210 and autonomously polls its verdict mailbox.

Input	Size (MiB)	FPGA median / p95 (ms)	GPU median / p95 (ms)	FPGA speedup
R1	5.28	2.679 / 2.680	1.327 / 1.409	0.50×
R2	11.45	3.074 / 3.075	2.162 / 2.328	0.70×
R3	18.00	2.919 / 2.925	3.077 / 3.093	1.05×
R4	28.92	2.885 / 2.886	4.618 / 4.678	1.60×
R5	32.58	2.935 / 2.939	5.138 / 5.178	1.75×
R6	39.37	2.808 / 2.811	6.081 / 6.198	2.17×
R7	46.29	2.695 / 2.698	7.058 / 7.094	2.62×
R8	53.10	2.750 / 2.753	8.012 / 8.067	2.91×

Read the speedup in either direction: values below 1× favor GPU_P2P. Selective HBM reads reduce traffic but do not guarantee lower latency than sequential delivery. The new measurements show a crossover, rather than a universal FPGA win. Compilation and cold-start costs are not included.

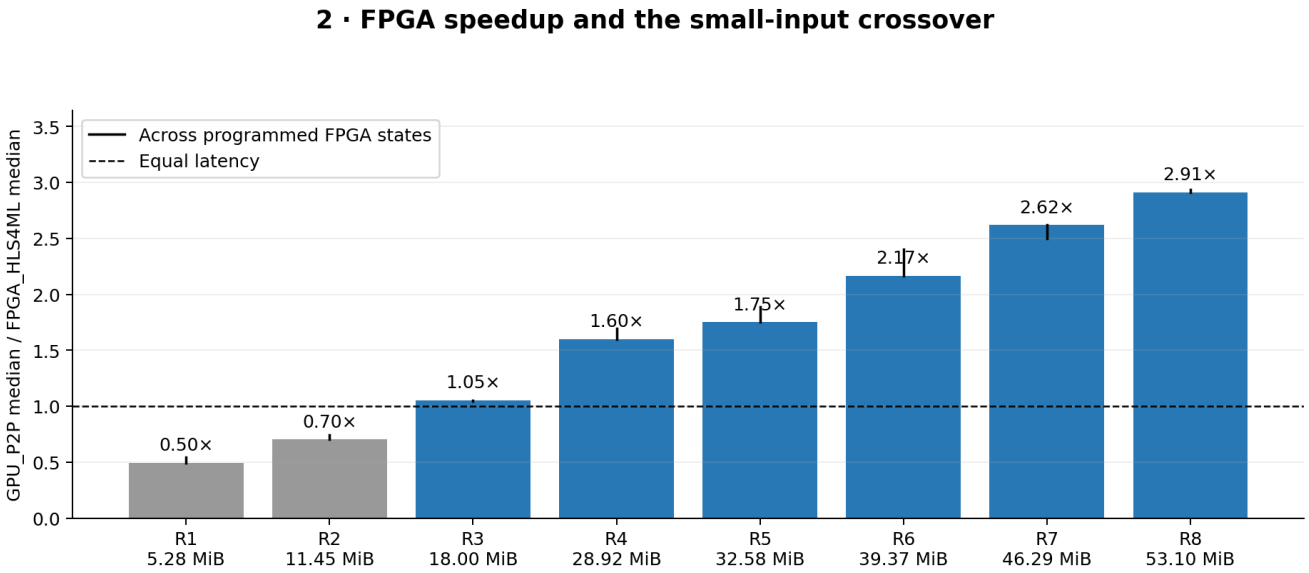
Reproducibility limitation: identical-image reprogramming changed some FPGA memory-delivery medians by more than 5%. Main bars retain the initial run; additional markers and Appendix A5 expose the other states. Within-program repeatability and exact classification checks passed.

1 · Verdict latency with the bitstream already in HBM



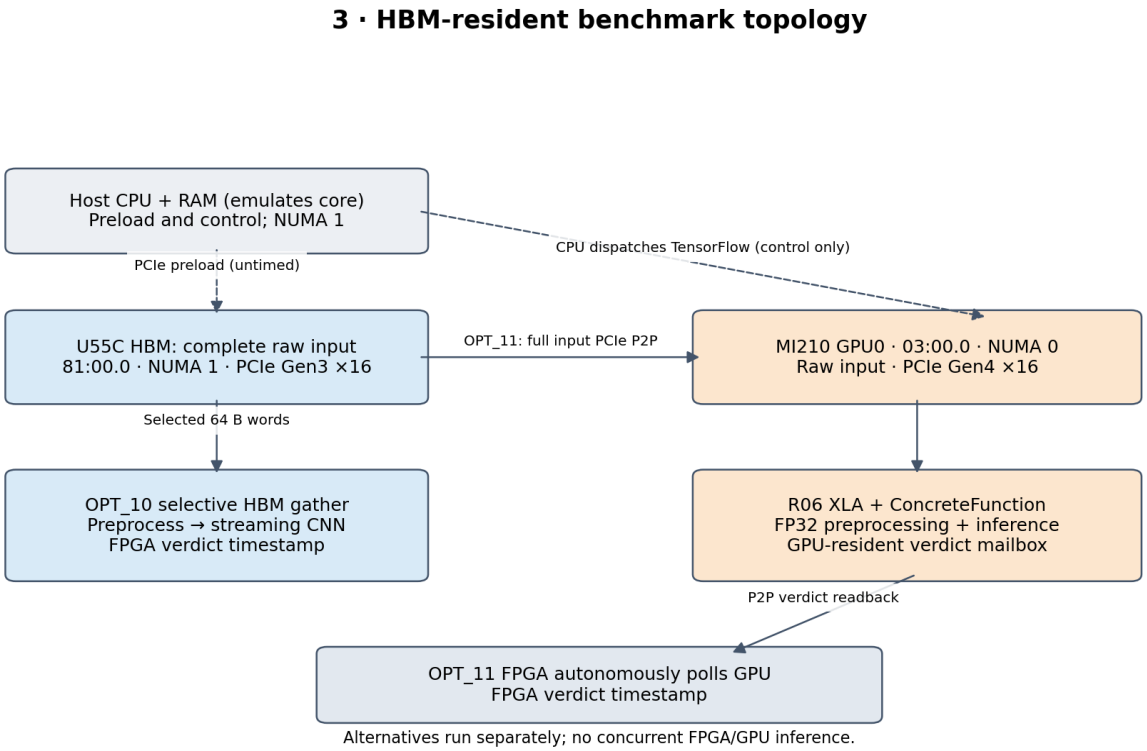
Bars: medians; black ticks: p95; 200 measured trials per workload after 20 warmups. The origin is accepted FPGA start, and the endpoint is the FPGA verdict event. FPGA bars use the initial program; hollow circles/diamonds show medians after two further identical-image programming cycles (200 and 20 trials). The initial-to-closing 5% stability gate failed for several workloads; this initialization sensitivity is retained and investigated in Appendix A5. HBM preload, file access, GPU compilation and partial reconfiguration are excluded. FPGA_HLS4ML is OPT_10 (quantized OPT_6 CNN); GPU_P2P is OPT_11 with R06 XLA compilation + ConcreteFunction FP32.

2 · FPGA speedup and the small-input crossover



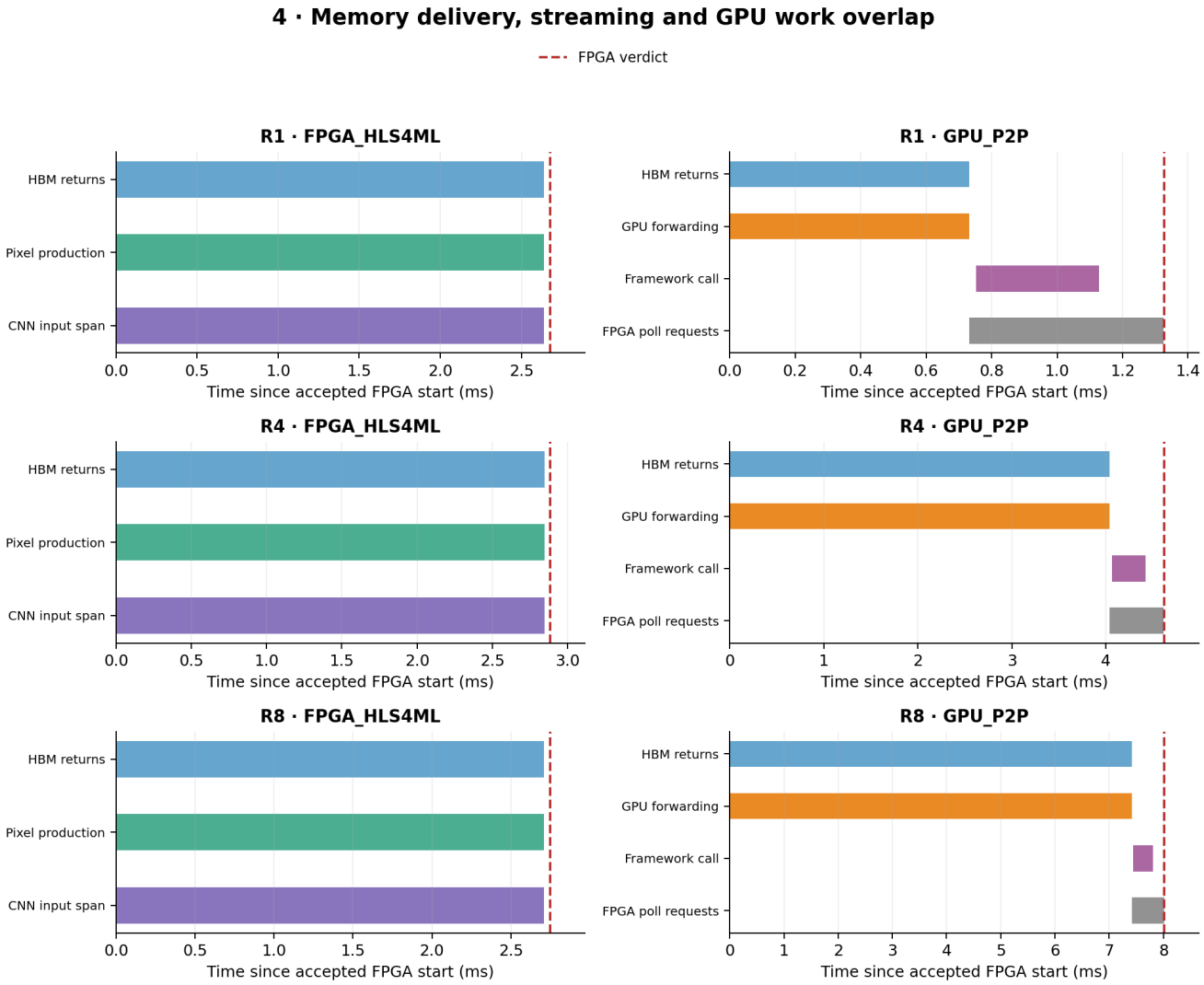
Above 1×: FPGA_HLS4ML is faster. Below 1×: GPU_P2P is faster. The FPGA requests only the words containing selected pixels, but those small reads have substantial overhead. The GPU forwards all bytes; its latency increases with input size. Thin vertical ranges show the speedup across three programmed FPGA states, not confidence intervals. Bar labels use the initial state.

3 · HBM-resident benchmark topology



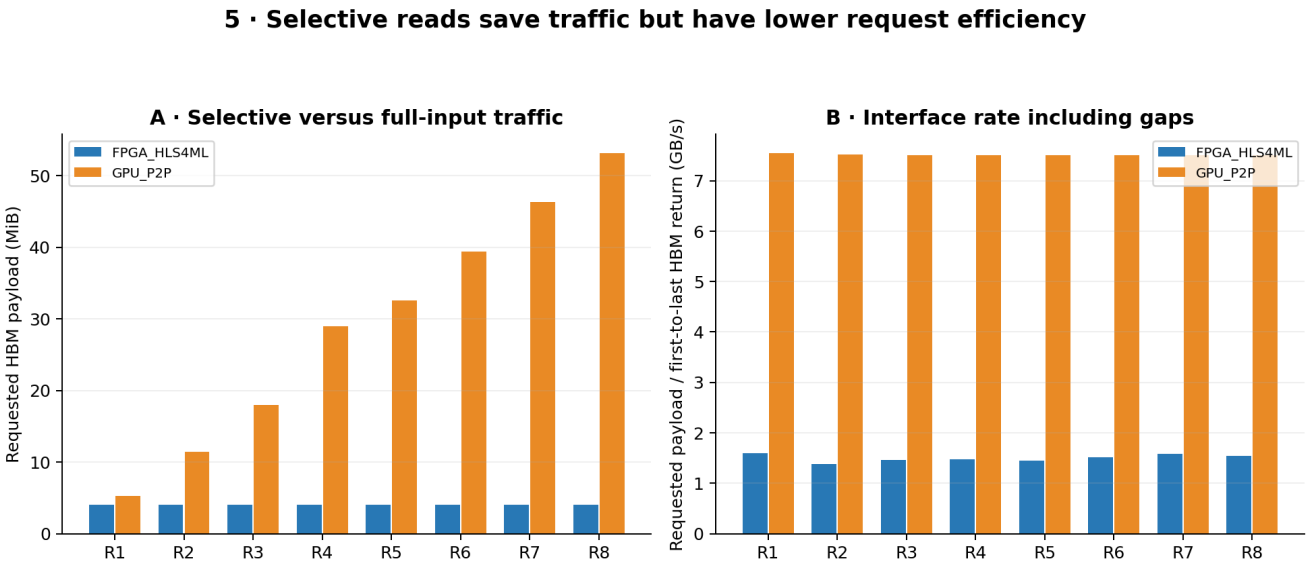
The host places the raw bitstream in HBM before measurement, emulating an integrated or soft CPU. The experiment does not implement an embedded CPU, PR, or fully hostless GPU execution. GPU inference remains CPU-dispatched. P2P is between devices attached to different NUMA nodes; the drawing shows the logical route, not a verified PCIe-switch diagram. U55C serial: XFL13BA4HVZYA.

4 · Memory delivery, streaming and GPU work overlap



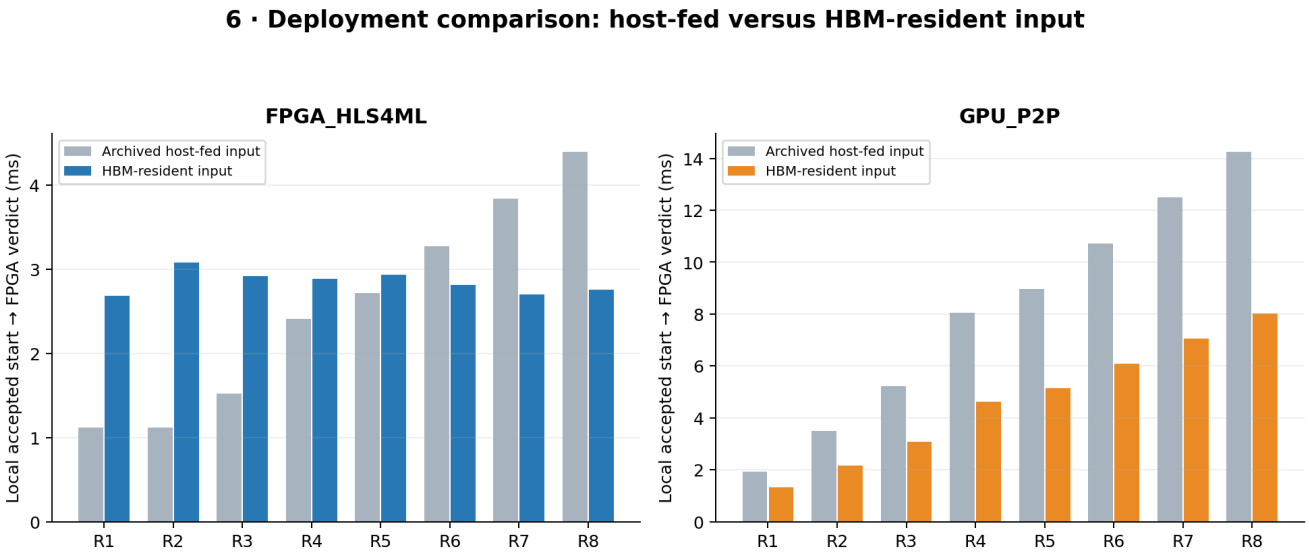
Each panel is one measured trial nearest its workload median; panels use their own x-axis limits. HBM, pixel and CNN spans overlap and must not be added. The CNN lane shows first-to-last input consumption, not compute-only latency. Its output tail ends at the dashed verdict line. GPU framework call includes dispatch, fused preprocessing/inference and synchronization; it is not a pure kernel span. GPU host-clock events use the affine MMIO calibration; FPGA events share the local counter. Polling overlaps waiting for GPU work, rather than adding the entire polling window after it.

5 · Selective reads save traffic but have lower request efficiency



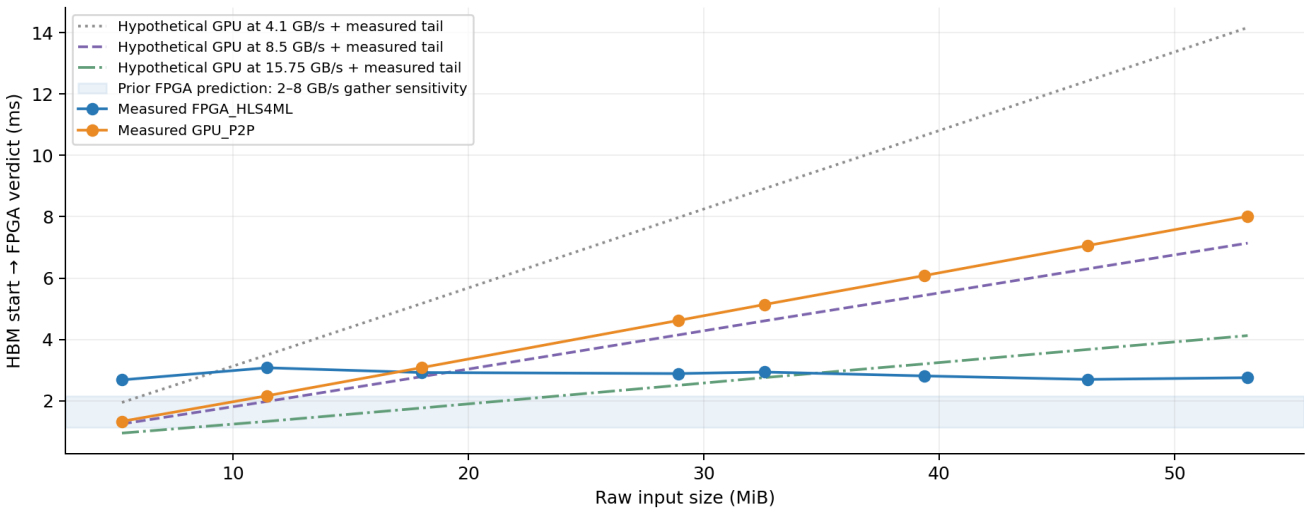
All eight real inputs use 65,536 distinct aligned 64-byte requests for FPGA_HLS4ML: 4 MiB requested for 64 KiB useful sampled bytes. GPU_P2P requests and forwards the complete raw input. The plotted rates include gaps within the first-to-last return interval and are not peak HBM bandwidth. Requested/interface payload is distinct from physical HBM-wire traffic, which is not measured. PCIe payload for OPT_10 is zero during checking; OPT_11 additionally sends all raw bytes to the GPU plus mailbox reads.

6 · Deployment comparison: host-fed versus HBM-resident input



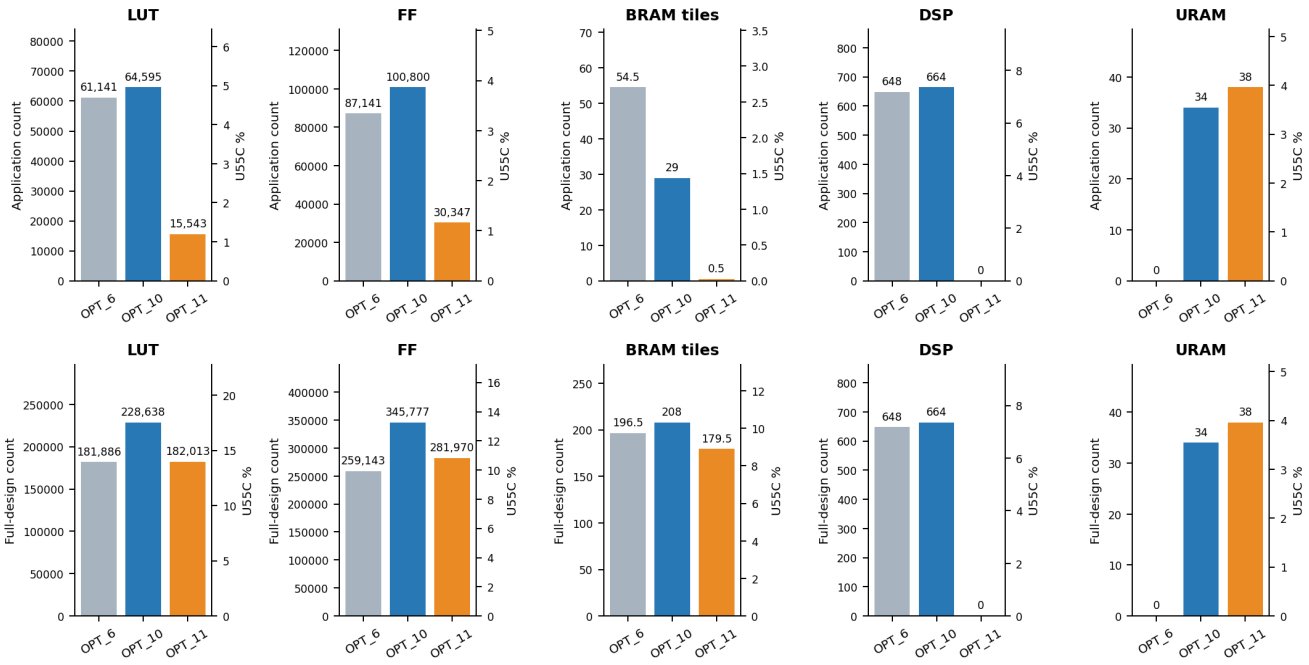
Archived measurements: 2026-10-21 R06 report; new measurements: 2026-11-08. Both use FPGA-local accepted start to verdict, but they begin with different input residency and use different transport wrappers. This is a deployment comparison, not an isolated causal estimate of HBM acceleration. The gather implementation can be slower than the old streaming checker at small sizes; the new GPU forwarding path benefits from direct HBM sourcing. Host/file-to-verdict scopes are deliberately excluded.

7 · Measured results and hypothetical P2P scenarios



Scenario curves replace the complete-input transfer with raw bytes / assumed rate, then add the new measured per-case tail from final GPU-write completion to FPGA verdict. They assume negligible startup and enough HBM supply bandwidth. Rates 4.1, 8.5 and 15.75 GB/s are scenarios, not achieved rates; 15.75 GB/s is especially optimistic because protocol overhead prevents sustaining the encoded-link ceiling. These curves are neither confidence intervals nor guarantees. The earlier 1.12 ms FPGA prediction was too optimistic for this gather implementation; the measured selective-read delivery time explains the difference.

8 · Routed resource costs: application and full design

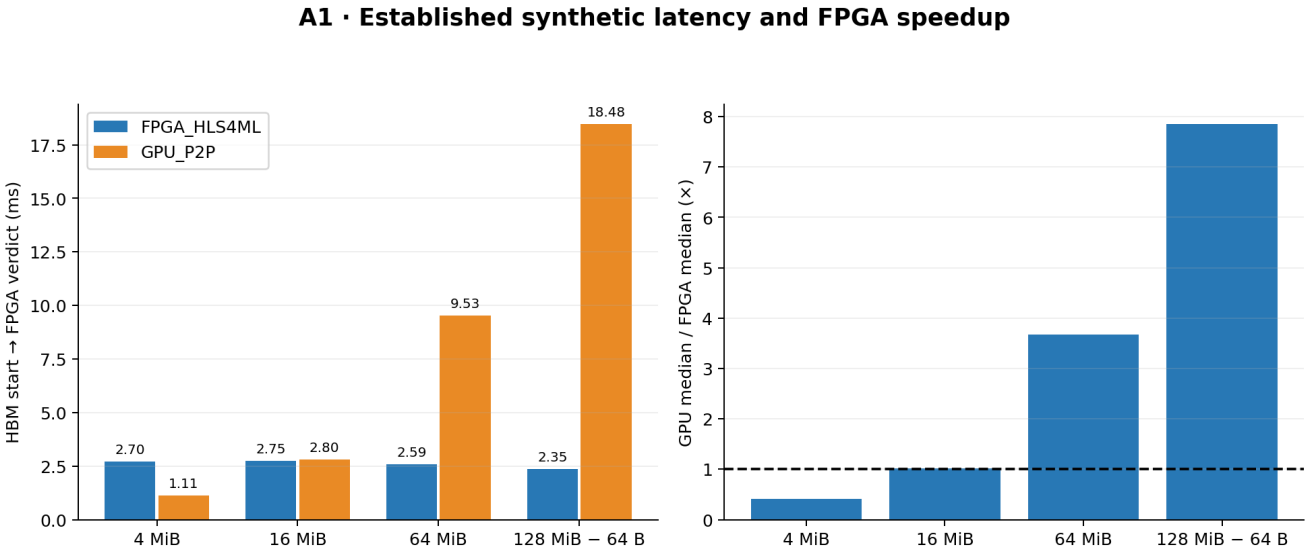


Top: inst_user_wrapper_0 including checker/transport logic. Bottom: complete routed design including the shell. Counts at left; U55C capacity utilization at right. BRAM tiles = RAMB36 + 0.5 × RAMB18. OPT_6 is the archived PCIe-streaming checker; OPT_10/11 enable HBM and contain different shell logic. Wrapper attribution is reported directly; no unmatched no-op baseline is subtracted. The GPU model uses external MI210 resources, which are not counted as FPGA area.

Appendix · Synthetic workloads and diagnostics

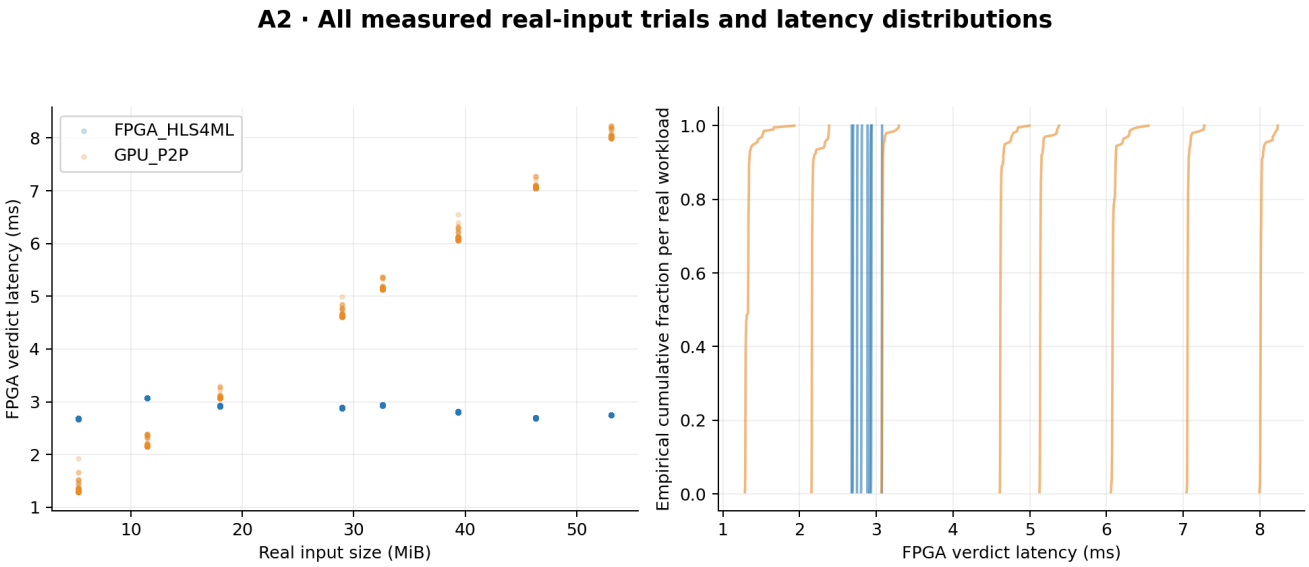
The primary comparison uses the same eight real inputs as the earlier report. Established synthetic sizes and preprocessing boundaries are supplemental. Synthetic byte patterns differ from real files; equal sizes alone do not imply equal request scheduling or identical model outputs.

A1 · Established synthetic latency and FPGA speedup



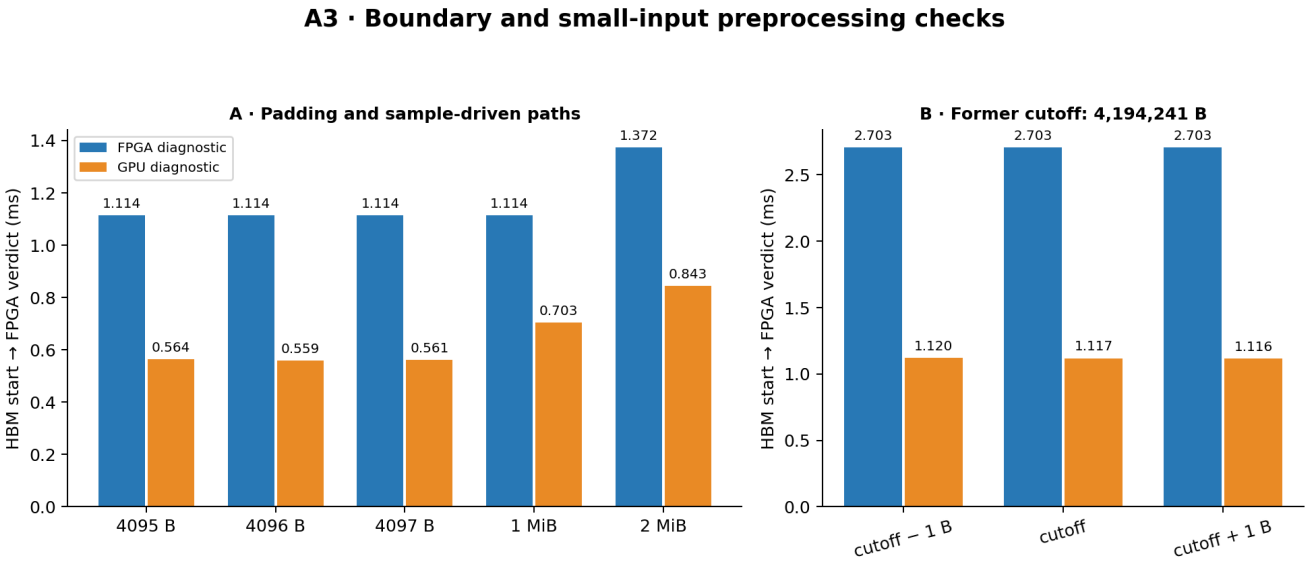
Same measurement scope as the real-input figures. The largest case remains 128 MiB – 64 B to retain the established GPU transport allocation convention. The 16 MiB point is shown without substituting a different test pattern. No old file-to-verdict result is mixed into this HBM-resident experiment.

A2 · All measured real-input trials and latency distributions



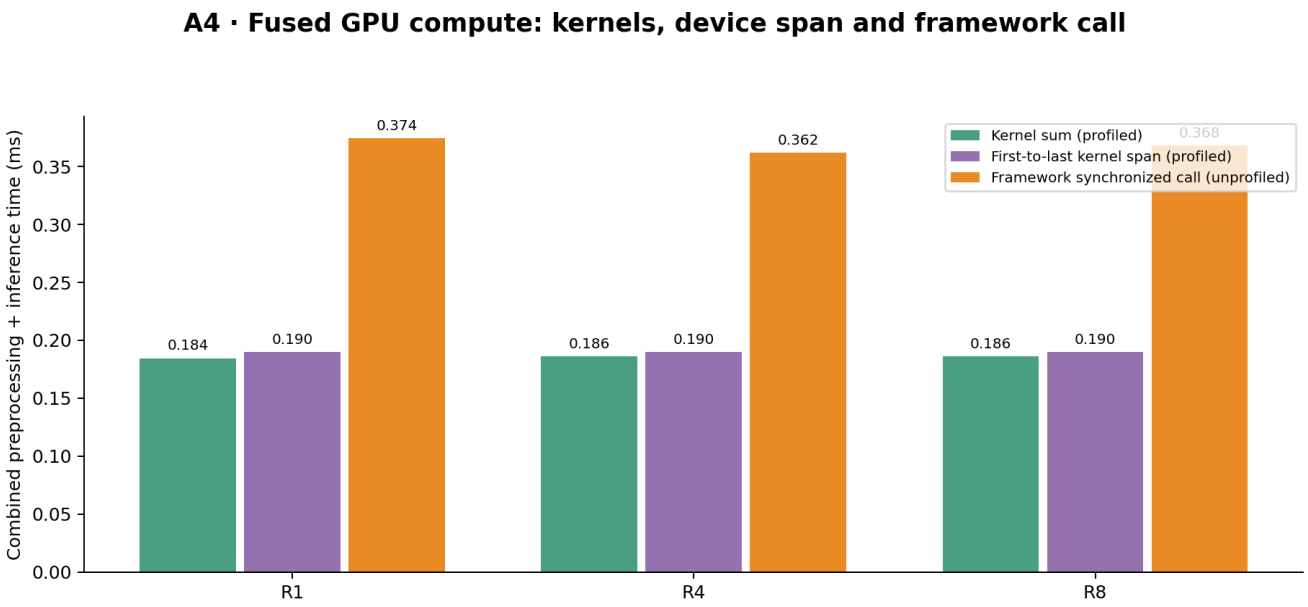
Each size contributes 200 unprofiled trials per variant. ECDFs are separate workload distributions, not pooled confidence intervals. Blue: FPGA_HLS4ML; orange: GPU_P2P.

A3 · Boundary and small-input preprocessing checks



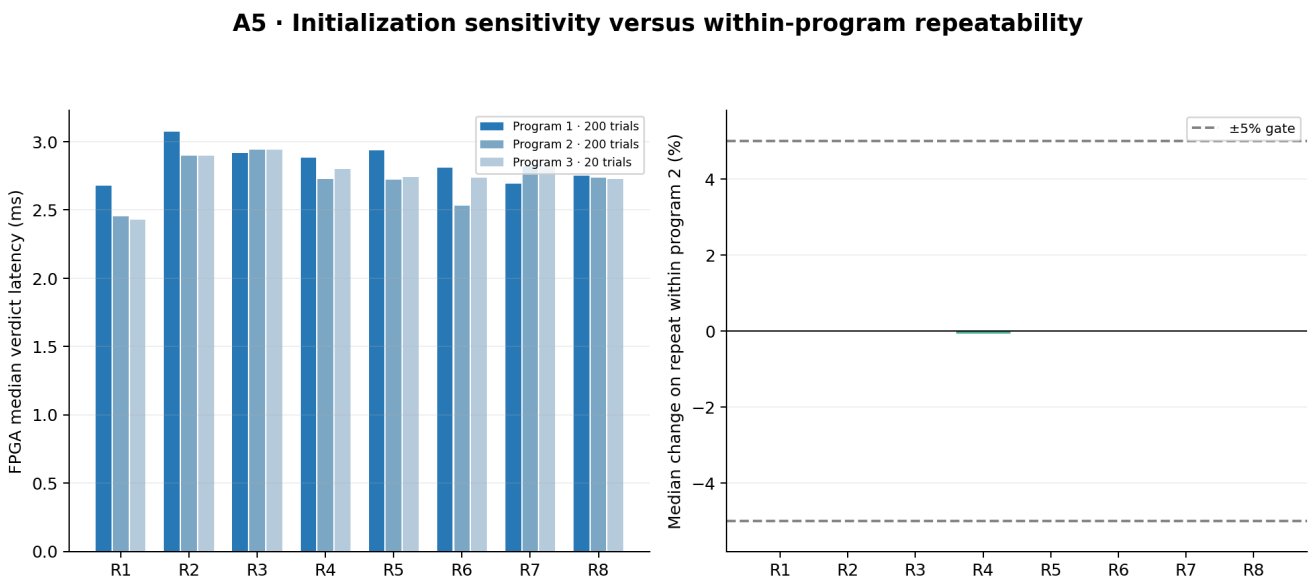
Panel A: 20 measured calls per case after two warmups. These cases use newly pinned deterministic random patterns and exact hls4ml CPU references; they are separate from the primary input set. Panel B: 200 measured calls after 20 warmups for each existing cutoff pattern; every call verifies parity. The new gather function generates 65,536 sampled/padded pixels directly rather than scanning every raw byte below the former cutoff. No matched old HLS-cycle rerun was available in this board campaign, so no isolated small-input speedup is claimed.

A4 · Fused GPU compute: kernels, device span and framework call



Kernel sum adds recorded kernel durations; device span runs from first kernel start to last kernel end, including intervening device gaps. Both are medians of five separate profiled calls. The synchronized framework call is a median of 200 unprofiled hardware-path calls and additionally includes dispatch and synchronization overhead. These are different scopes and trial sets. XLA fuses preprocessing with model operations, so no artificial preprocessing/inference split is shown. Transport, mailbox publication and FPGA polling are outside the compute marker. Profiling starts after one validated P2P delivery and drained operation; it does not call the HIP mailbox adapter while tracing. This isolates a ROCm profiler/publication interoperability failure encountered in the first diagnostic attempt, which is retained in the campaign logs and excluded from reported measurements.

A5 · Initialization sensitivity versus within-program repeatability



Three programming cycles use the identical OPT_10 image and driver. Program 1 and 2 each contain 200 measured calls after 20 warmups per workload; program 3 contains 20 measured calls after two warmups. A second 200-call run in program 2 remains stable. The older transport adapter gives the program-2 result too, ruling out the added start-acceptance guard. The shift occurs in the first-to-last HBM return interval; the CNN tail and exact outputs remain stable. Physical placement logs were captured, but clock/PHY initialization sensitivity remains a hypothesis. This limitation must be retained when citing a single median from the main plots.

Validation, timing and reproducibility

Gate	Evidence / outcome
opt10_admission	passed; 520 validated calls
opt11_admission	passed; 520 validated calls
opt11_stale_visibility	passed; 26 validated calls
opt10_diagnostics	passed; 110 validated calls
opt11_diagnostics	passed; 110 validated calls

Opening/closing control: maximum absolute median change 9.825%; the 5% cross-program stability gate failed. Correctness, byte counts and clock calibration passed. A second 200-trial repeat without reprogramming changed medians by at most 0.076%, and the old adapter reproduced the same second-program state. This localizes the change to memory delivery associated with the programmed state; it does not prove a specific HBM calibration or clock-phase cause. The report shows all three programmed states and does not claim cross-program stability. A single board lease serialized the runs; no new FPGA build was required.

The affine clock fits use 32 MMIO brackets before and after each workload. Fitted tick bounds span 3.999921–3.999997 ns. Primary intervals use one FPGA clock, avoiding host-observation delay. GPU host-clock annotations carry the alignment uncertainty recorded in data/trial_events.json. The ABI5 signature, status, trial, event-valid bits, counts and final drain are checked on every recorded call.

OPT_10 admission requires exact quantized score bits and 65,536 pixels. OPT_11 requires finite scores within 1e-4 absolute + 1e-4 relative error of its FP32 CPU model, identical threshold verdict, exact raw-byte visibility, and completed forwarding counts. The two models have different precisions; their logits are not required to match each other. HBM is verified before and after each workload and re-offloaded after readback, because LOCAL_SYNC restores host-access mappings.

Unavailable counters: physical HBM-wire traffic, maximum outstanding requests, aggregate HBM stalls and FIFO maximum occupancy. OPT_10 exposes final/current FIFO occupancy only; zero at completion must not be interpreted as zero peak. GPU kernel sum and device span come from separate profiled calls; unprofiled timing is used for all main latency figures. Fused XLA work is not split into invented preprocessing and CNN-only components.

Reproduction: run the maintained hbm/benchmark.py under the board controller and matching driver, then hbm/analyze.py followed by hbm/report.py. Raw trials, calibration, commands, source snapshots and per-stage logs are in hls4ml/optimize/runs/hbm_benchmark_20261108. This folder preserves the old report and uses its input identities, resource conventions, clock model and archived data.